

# Day 5 Practical: Build a Home Wi-Fi Network in Cisco Packet Tracer

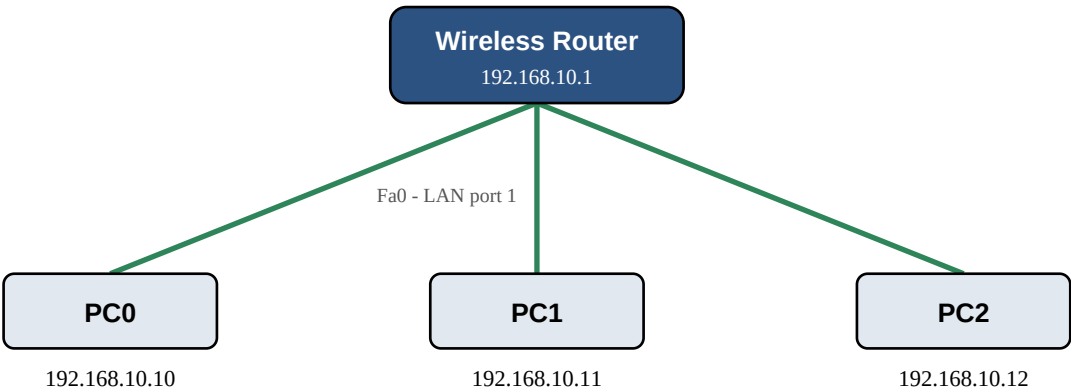
**Project file:** HomeWiFiNetwork.pkt | **Duration:** 1 hour | **Tool:** Cisco Packet Tracer

**Goal:** Create a small home network with 3 PCs and 1 wireless router, assign static IPs on the same subnet, set the default gateway, test connectivity with ping, and document everything.

## Addressing Plan

| Device          | Interface                  | IP Address    | Subnet Mask   | Default Gateway |
|-----------------|----------------------------|---------------|---------------|-----------------|
| Wireless Router | LAN (Internet side unused) | 192.168.10.1  | 255.255.255.0 | N/A             |
| PC0             | FastEthernet0              | 192.168.10.10 | 255.255.255.0 | 192.168.10.1    |
| PC1             | FastEthernet0              | 192.168.10.11 | 255.255.255.0 | 192.168.10.1    |
| PC2             | FastEthernet0              | 192.168.10.12 | 255.255.255.0 | 192.168.10.1    |

Topology: HomeWiFiNetwork.pkt (192.168.10.0/24)



## Step 1: Create the Project and Build the Topology

- Open Cisco Packet Tracer and choose **File > New**.
- In the device panel (bottom left), select **End Devices** and drag **3 PCs** onto the workspace. They are named PC0, PC1, PC2.
- Select **Network Devices > Wireless Devices** and drag one **Wireless Router** (e.g., Linksys WRT300N) to the workspace.
- Select **Connections** and choose **Copper Straight-Through** cable. Click a PC, select **FastEthernet0**, then click the router and select an Ethernet port (Ethernet 1, 2, 3). Repeat for all three PCs.
- Wait until all link lights turn **green** (this can take a few seconds).
- Save with **File > Save As** and name it **HomeWiFiNetwork.pkt**.

Tip: Use a straight-through cable because PCs connect to a router/switch (different device types). Wireless is optional here; wired links are the simplest way to make the connection reliable.

**Screenshot 1:** Insert a screenshot of the completed topology here.

## Step 2: Set the Router LAN IP (192.168.10.1)

- Click the router and open the **GUI** tab.
- Go to **Setup > Basic Setup**. Under **Network Setup**, set Router IP to **192.168.10.1** and Subnet Mask to **255.255.255.0**.
- Set **DHCP Server** to **Disabled** (we are using static addresses).
- Scroll down and click **Save Settings**.

**Screenshot 2:** Insert a screenshot of the router Basic Setup page here.

## Step 3: Assign Static IPs on Each PC

- Click the PC and open the **Desktop** tab, then **IP Configuration**.
- Select **Static** (not DHCP).
- Enter the values from the addressing table: IPv4 address, subnet mask **255.255.255.0**, and default gateway **192.168.10.1**.
- Close the window. The settings save automatically.
- Repeat for PC0 (.10), PC1 (.11) and PC2 (.12).

**Screenshots 3, 4, 5:** Insert one IP Configuration screenshot per PC (PC0, PC1, PC2).

## Step 4: Test Connectivity with Ping

On each PC open **Desktop > Command Prompt** and run the pings below.

| From | Ping targets                               | Expected result  |
|------|--------------------------------------------|------------------|
| PC0  | 192.168.10.11, 192.168.10.12, 192.168.10.1 | Replies from all |
| PC1  | 192.168.10.10, 192.168.10.12, 192.168.10.1 | Replies from all |
| PC2  | 192.168.10.10, 192.168.10.11, 192.168.10.1 | Replies from all |

Example (from PC0):

```
C:\> ping 192.168.10.11
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Ping statistics: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

The first ping may time out once while ARP resolves; this is normal. A successful result shows **0% loss**.

**Screenshot 6:** Insert a Command Prompt screenshot showing successful pings.

## Troubleshooting

- **Red link lights:** wait a few seconds, or check the cable type and ports.
- **Request timed out:** verify all devices are in 192.168.10.x with mask 255.255.255.0 and that there are no typos.

- **Cannot ping the router:** confirm you clicked **Save Settings** in the router GUI.
- **Duplicate IP:** make sure each device has a unique address.

## Step 5: Documentation Checklist

| # | Item                                                          | Done |
|---|---------------------------------------------------------------|------|
| 1 | Screenshot of topology (3 PCs + wireless router, green links) | [ ]  |
| 2 | Screenshot of router LAN settings                             | [ ]  |
| 3 | IP Configuration screenshots for PC0, PC1, PC2                | [ ]  |
| 4 | Ping test screenshots (PC to PC and PC to router)             | [ ]  |
| 5 | HomeWiFiNetwork.pkt saved                                     | [ ]  |

**Conclusion:** All three PCs share the 192.168.10.0/24 subnet, use the router (192.168.10.1) as their default gateway, and can communicate with each other and with the router, confirming the home network is configured correctly.